$$
\begin{aligned}
\langle \hat{q}_i(\tau)\hat{q}_i \rangle_{\hat{\rho}\mathcal{E}} &= \frac{1}{2m_i\omega_i} \left\{ [1 + N_i(T)]\, \mathrm{e}^{-\mathrm{i}\omega_i\tau} + N_i(T)\mathrm{e}^{\mathrm{i}\omega_i\tau} \right\} \\
&= \frac{1}{2m_i\omega_i} \left\{ [1 + 2N_i(T)] \cos(\omega_i\tau) - \mathrm{i} \sin(\omega_i\tau) \right\} \\
&= \frac{1}{2m_i\omega_i} \left\{ \coth\left( \frac{\omega_i}{2k_\mathrm{B}T} \right) \cos(\omega_i\tau) - \mathrm{i} \sin(\omega_i\tau) \right\}, \qquad (5.13)
\end{aligned}
$$